4. Monday, April 7. Logically equivalent statements (2.2), Sets (2.3)

Last Time: · Tautologies and contradictions
· Logically equivalent statements.

Examples 1) $\neg(\neg P) \equiv P$

2) $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

3) $P \to Q \equiv \neg P \vee Q$

Example 4: (Negations of conditionals)
We have $\neg(P \to Q) \equiv P \wedge \neg Q$ for any statements P and Q

Proof: We use the previous examples:

$$\neg(P \to Q) \overset{(3)}{\equiv} \neg(\neg P \vee Q) \overset{(2)}{\equiv} \neg(\neg P) \wedge \neg Q \overset{(1)}{\equiv} P \wedge \neg Q$$

Therefore $\neg(P \to Q) \equiv P \wedge \neg Q$ □

More on Conditionals: For a conditional $P \to Q$, we have three related conditionals

- The converse $Q \to P$
- The inverse $\neg P \to \neg Q$
- The contrapositive $\neg Q \to \neg P$

Next we compute the truth tables

P	Q	$\neg P$	$\neg Q$	$P \to Q$	$Q \to P$	$\neg P \to \neg Q$	$\neg Q \to \neg P$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	F	T	F	T	T
F	F	T	T	T	T	T	T

Therefore $P \to Q \equiv \neg Q \to \neg P$!

This also justifies the Truth Table for $P \to Q$.

Next we summarize some useful logical equivalences

Theorem 5: Let P, Q and R be statements. Then

- $\neg(\neg P) \equiv P$
- $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$ } DeMorgan's Laws
- $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
- $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$ } Distributive Laws
- $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$

- $P \to Q \equiv \neg P \vee Q$
- $\neg(P \to Q) \equiv P \wedge \neg Q$
- $P \to Q \equiv \neg Q \to \neg P$
- $P \leftrightarrow Q \equiv (P \to Q) \wedge (Q \to P)$

Sets

Definition: A **set** is a well-defined collection of objects.

The objects in a set are called the **elements** of the set.

Remark: To understand a set, we just need to know what is and is not an element of the set.

Notation: Let S denote a set

We write
- $x \in S$ to say x is an element in S — $\in$ = \in
- $x \notin S$ to say x is not an element in S — $\notin$ = \notin

Examples:
- $\{a, b, c\}$ ← A set with three elements: a, b, c
- $\{b, c, a\}$ ← The same set.
- $\{a, b, d\}$ ← A different set.
- $\{a, b, c, a\} = \{a, b, c\}$ same set.

In this example we wrote sets using the roster method. This means we listed all elements within curly braces.

If there are too many elements to write but they follow a pattern, we can also write some elements and indicate the pattern continues with ...

\mathbb{N}

Example 8 • $\mathbb{N} = \{1, 2, 3, \dots\}$ all natural numbers

• $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ all integers "Z" stands for zahlen, number in German.

So $-2 \in \mathbb{Z}$ but $-2 \notin \mathbb{N}$.

Definition 9 The cardinality of a set is the number of elements in that set.

Example 10 • $\{a, b, c\}$ has cardinality 3. Same for $\{a, b, c, a\}$

• $\mathbb{N}$ has infinite cardinality.